

Goal

M classical clients jointly delegate a public quantum circuit on their combined private inputs to two entangled, non-communicating quantum provers PV, PP such that all inputs stay secret and the result is always verified.

“Should we trust IBM, Amazon, Atom and other cloud quantum providers to actually run the circuits we send them?”

Why two provers?

Entanglement-based VBQC (Reichardt–Unger–Vazirani, CGJV) lets a fully *classical* verifier information-theoretically certify a remote computation via CHSH rigidity/self-testing.

The multi-client problem

Naively running classical MPC oracle inside a single-client delegation protocol is *insecure*: Bartusek showed a colluding client+server pair can leak other clients’ inputs via *input-dependent aborts*.

Our contribution

- Modify CGJV’s Leash protocol into **Leash’**: separate the accept/reject test outcome from the computation outcome; retains completeness & soundness.
- Prove **simulator-based blind verifiability** via the abstract/constructive cryptography (AC) framework and an adapted two-prover DFPR local-reduction chain.
- Extend Leash’ to M clients with classical MPC: adds **input privacy between clients**, even against a coalition colluding with one (not both) prover.

Physical prover separation replaces the LWE-trapdoor machinery that Mahadev-style single-server protocols need to block input-dependent aborts, giving a simpler and more efficient multi-client VBQC protocol.

Protocol	Provers	Rounds
RUV 2012	2	$O \geq g^{8192}$
McKague 2013	$\text{poly}(n)$	$\geq 2^{153} g^{22}$
GKW 2015	2	$O(g^{2048})$
HPDF 2015	$\text{poly}(n)$	$O(g^4 \log g)$
Mahadev 2018	1	$O(\text{constant})$
CGJV 2024	2	$O(g \log g)$
Leash’ (multi-client)	2	$O(4g \log g)$

Abstract/Constructive Cryptography

A protocol π uses a *resource* $\mathcal{R}$ to *construct* an ideal resource $\mathcal{S}$ (Maurer–Renner AC framework); constructions compose sequentially and in parallel. This is a top-down, simulator-based security definition with lower levels inheriting properties of higher levels. The ideal resource is a *specification* of the desired functionality, and the real resource is a *concrete implementation* of it. The distinguisher $\mathcal{D}$ models an adversary that can interact with either the real or ideal world.

A resource has interfaces $\mathcal{I} = \{A, B, C, \dots\}$, one per party. A **filter** $\perp$ restricts an interface to honest use. A **converter** π_i plugs into one interface. A **distinguisher** $\mathcal{D}$ connects to every interface, giving

$$d(\mathcal{R}, \mathcal{S}) = \max_{\mathcal{D}} \Pr[\mathcal{D}\mathcal{R} = 1] - \Pr[\mathcal{D}\mathcal{S} = 1].$$

A construction is secure if **simulators** σ_i at dishonest interfaces make the ideal world indistinguishable from the real one:

$$d(\pi\mathcal{R}, \mathcal{S}\sigma) \leq \epsilon, \quad d(\pi\mathcal{R}, \mathcal{S}\perp) = 0.$$

We prove this property twice, for single-client Leash’ (see *Ideal and Real Resources*), and again for the M -client MPC-composed protocol.

Quantum One-Time Pad

Much like the classical one-time pad, the quantum one-time pad (QOTP) encrypts a qubit $|\psi\rangle$ with two random bits (a, b) :

$$|\psi\rangle \mapsto X^a Z^b |\psi\rangle.$$

The QOTP is information-theoretically secure: the ciphertext is a maximally mixed state,

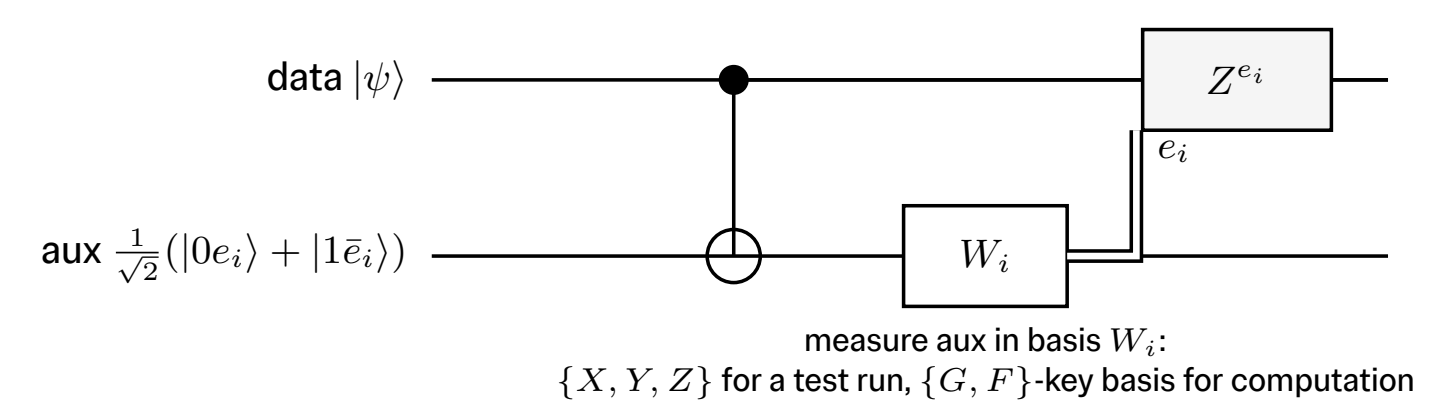
$$\frac{1}{4} \sum_{a,b \in \{0,1\}} X^a Z^b |\psi\rangle\langle\psi| Z^b X^a = \frac{\mathbb{I}}{2}.$$

Pauli gates commute through the QOTP up to a key update:

$$X^a Z^b X = X^{a \oplus 1} Z^b, \quad X^a Z^b Z = X^a Z^{b \oplus 1}.$$

The T-Gate Gadget

Broadbent’s linear-overhead EPR protocol delegates a Clifford+ T circuit to a prover holding EPR-pair halves; the verifier one-time-pads the input and picks uniformly among indistinguishable *computation*, *X-test*, and *Z-test* runs. Non-Clifford T gates need a **T -gadget**: a fresh EPR pair is consumed as a magic-state resource, teleporting T onto the data wire and producing an outcome used both for the key update and, in a test run, as a trap.



Reporting e_i lets the verifier (1) update the running one-time-pad keys (a_j, b_j) on target wire j , and (2) in an X/Z-test run, check e_i as a **trap**; mismatch flags err. Repeating once per logical T gate ($t = \sum_{\ell} t_{\ell}$ gates, d layers) gives the full delegation branch of Leash’.

Modified CGJV Wrapper

Leash’ round. Plays rigidity with probability p_r or delegation (modified Broadbent) with probability $1 - p_r$, returning $Y \in \{0, 1, \text{ok}, \text{err}\}$; the computation bit is *separated* from the test/trap accept-reject decision.

Sequential wrapper $W_L(K)$. Run K i.i.d. rounds; tally $M_c = \#\{y_i \in \{0, 1\}\}$ and error counts $E_{\text{comp}}, E_{\text{test}}, E_{\text{rigid}}$:

$$W_L(K) = \begin{cases} \text{err}, & E_{\text{comp}} > 0, \\ \text{err}, & E_{\text{test}} > \frac{(1-p_r)K}{96}, \\ \text{err}, & E_{\text{rigid}} > \frac{p_r \epsilon_0 K}{32}, \\ \text{err}, & M_c < \frac{5(1-p_r)K}{16}, \\ \text{maj}(M_0, M_1), & \text{otherwise.} \end{cases}$$

Theorem 1 (Completeness). $\Pr[W_L(K) = f(x) \mid \text{honest}] \geq 1 - e^{-\Theta(K)}$.

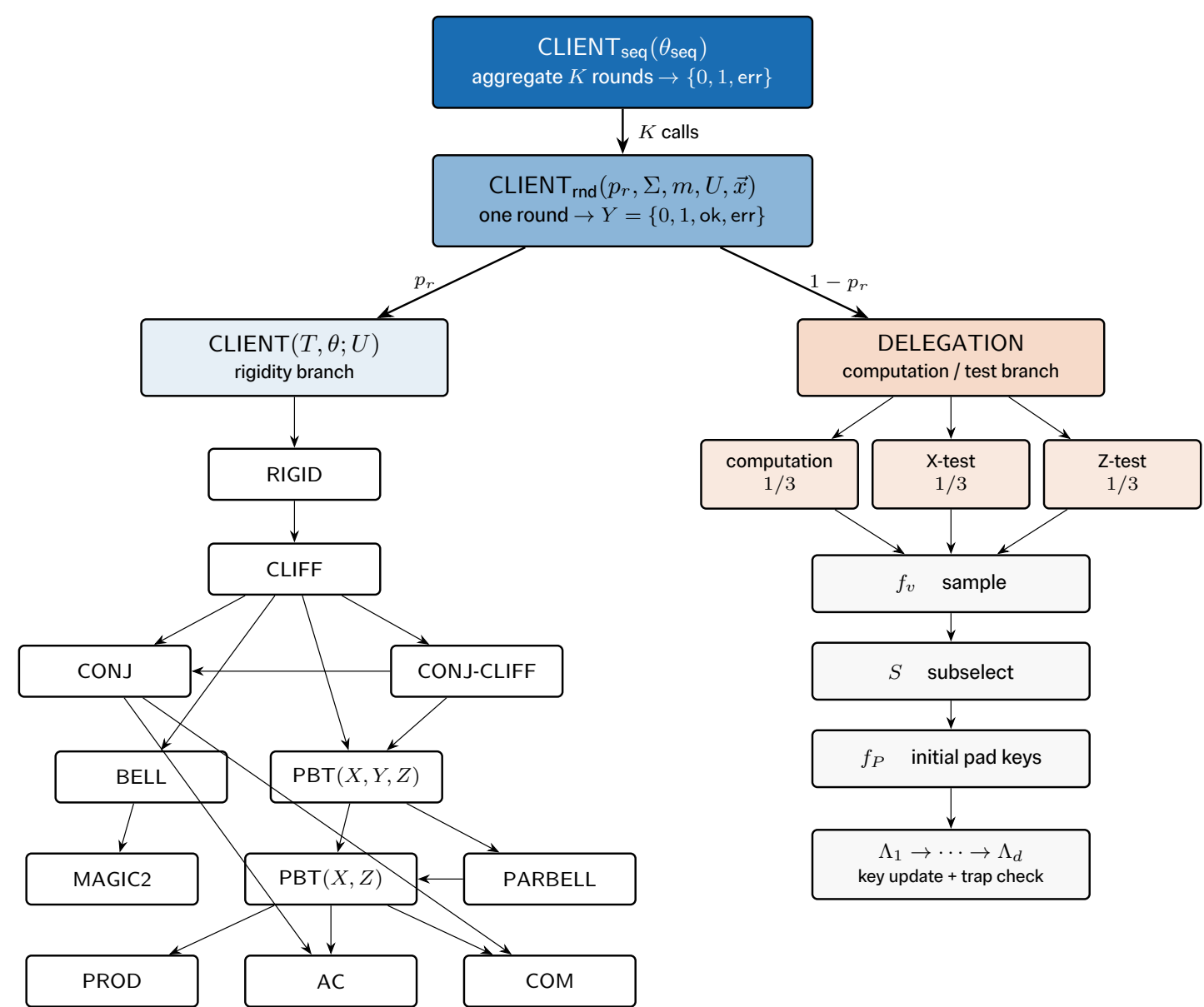
Theorem 2 (Soundness). *For any adaptive prover pair,*

$$\Pr[W_L(K) = 1 - f(x)] \leq e^{-(1-p_r)^2 K / (4608\gamma^2)}, \quad \gamma = 1 + \max\left\{1, \frac{1-p_r}{3\epsilon_0 p_r}\right\}.$$

Both proofs use an Azuma–Hoeffding supermartingale: signed per-round indicators (wrong output, trap failure, rigidity failure) sum to a supermartingale with expected increment ≤ 0 , so a wrong majority needs an exponentially unlikely deviation. The wrapper *decouples* the two error channels needed for the simulator construction.

Client Functions

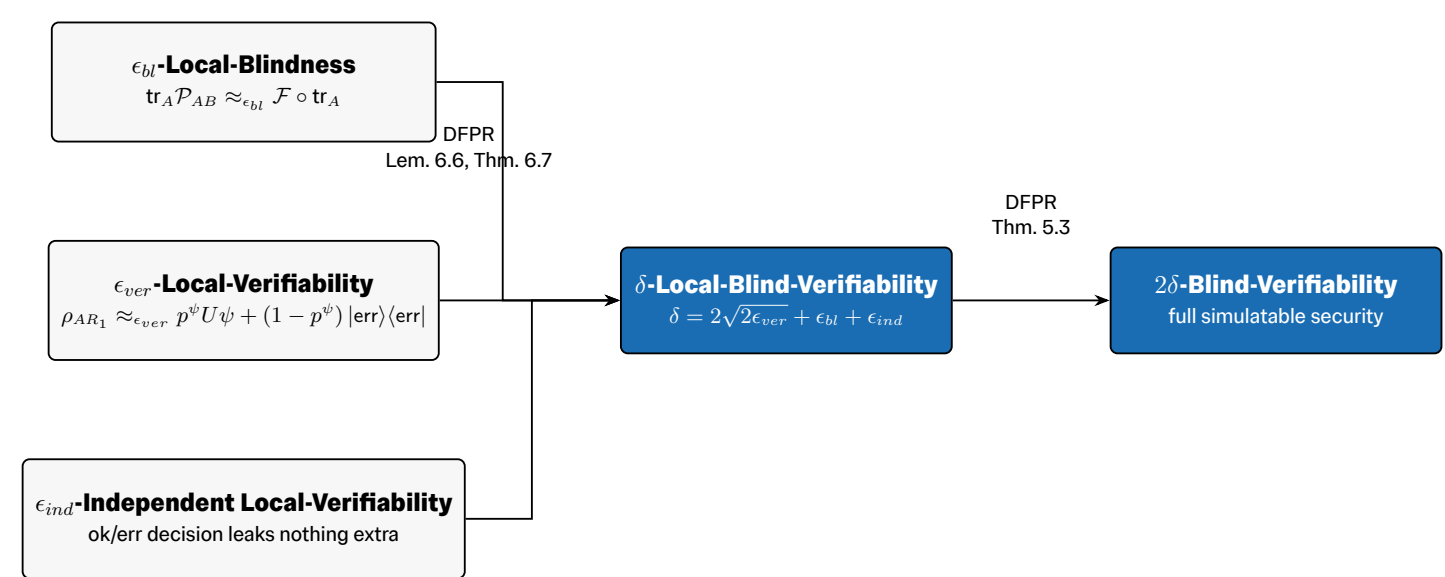
The verifier’s behavior is organized in three layers. Each round then splits into an explicit rigidity branch and the delegation branch used for the computation:



Rigidity leaves COM, PROD, MAGIC2, AC are the base consistency/product/magic-square/anticommutation self-tests. **Delegation** selects computation, X-test, or Z-test uniformly, then calls $f_v, S, f_P, \Lambda_1, \dots, \Lambda_d$ in order.

The DFPR Local-Reduction Chain

CGJV’s Leash protocol does not prove blindness *simulatably*. We follow **DFPR**: reduce simulatable blind verifiability to three *local* properties provable directly on the honest-Alice/dishonest-Bob map $\mathcal{P}_{AB}$.



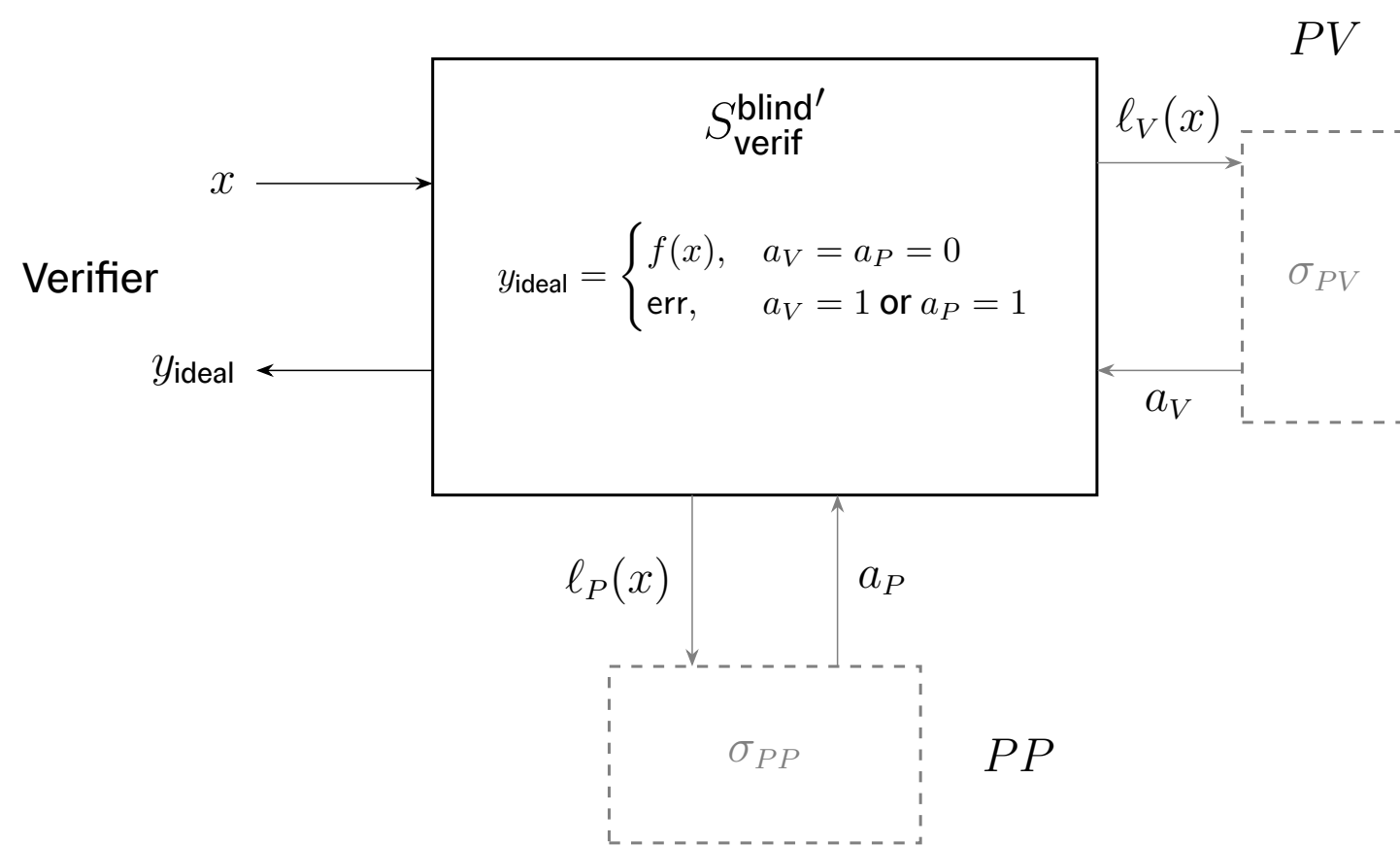
Key definitions. *Local-blindness*: Bob’s view is reproducible by an input-independent map $\mathcal{F}$. *Local-verifiability*: Alice’s output is, up to ϵ_{ver} , either correct $U|\psi\rangle$ or orthogonal $|\text{err}\rangle$. *Independence*: the ok/err decision itself leaks nothing beyond the permitted class.

Our adaptation. We instantiate this chain for the *two-prover, no-direct-communication* setting: absorbing the honest prover into the resource turns Leash’ into a two-interface DQC protocol, and local (in)dependence/blindness lemmas for both choices of malicious prover give simulators σ_{PV}, σ_{PP} for $\mathcal{S}_{\text{verif}}^{\text{blind}'}$.

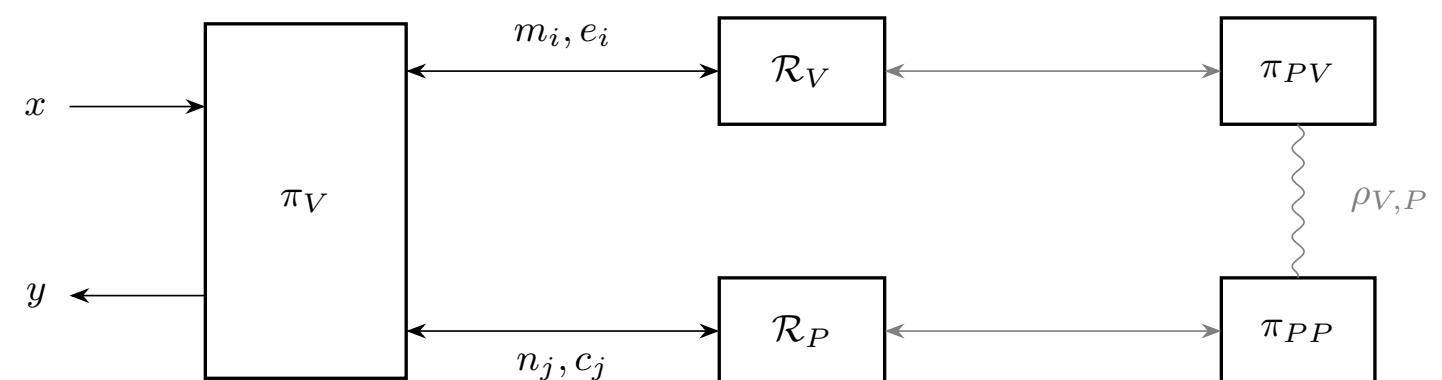
Ideal and Real Resources

The AC framework shows sequential Leash’ *constructs* a two-prover ideal resource with simulatable correctness and blindness.

Definition 1 (Modified two-prover ideal resource $\mathcal{S}_{\text{verif}}^{\text{blind}'}$). Receives x at the verifier interface; reveals $\ell_V(x)$ at PV , $\ell_P(x)$ at PP ; receives control bits a_V, a_P ; outputs $f(x)$ if $a_V = a_P = 0$, else err; never the incorrect bit $1 - f(x)$.



Ideal resource. $\mathcal{S}_{\text{verif}}^{\text{blind}'}$ with simulators σ_{PV}, σ_{PP} shown as dashed gray converters.



Real resources. π_V talks to non-communicating provers π_{PV}, π_{PP} sharing the entangled state $\rho_{V,P}$.

Definition 2 (Correctness).

$$d(\pi_V \mathcal{R} \pi_{PV}^H \pi_{PP}^H, \mathcal{S}_{\text{verif}}^{\text{blind}'} \perp_{PV} \perp_{PP}) \leq \epsilon_c.$$

For blindness the threat model exposes the verifier plus exactly one prover interface; we bound both cases separately.

Definition 3 (Blind-Verifiability).

$$\begin{aligned} d(\pi_V \mathcal{R} \pi_{PV}^H, \mathcal{S}_{\text{verif}}^{\text{blind}'} \perp_{PV}) &\leq \epsilon_{bv}, \\ d(\pi_V \mathcal{R} \pi_{PP}^H, \mathcal{S}_{\text{verif}}^{\text{blind}'} \perp_{PP}) &\leq \epsilon_{bv}. \end{aligned}$$

Threat Model

We consider collusion between a prover and a subset of clients, a case omitted in some previous work. Specifically,

$$\begin{aligned} \text{Clients: } \mathcal{C} &\equiv \{C_1, C_2, \dots, C_M\}, \\ \text{Provers: } \mathcal{P} &\equiv \{PV, PP\}, \\ \text{Colluding subset: } \mathcal{A} &\subseteq \mathcal{C}, \text{ such that } |\mathcal{A}| \leq t. \\ \text{Total coalition: } \mathcal{A} \cup \{PV\}, &\text{ or } \mathcal{A} \cup \{PP\}. \end{aligned}$$

To retain the no-communication assumption, a colluding client subset may communicate with either PV or PP, but not both. We also exclude two distinct colluding subsets and assume that at least one prover is honest. The case of two malicious non-communicating provers is left for future work.